

Workshop 11

Problem 1

Keywords: L'Hôpital's Rule

Compute the following limits. If you use L'Hôpital's rule, make sure to indicate where and say which indeterminate form allows you to use it.

(a) $\lim_{x \rightarrow \infty} \left(\frac{x^2}{x^2 + 2x + 1} \right)^x.$

(b) Find a value of the constant A for which the limit

$$\lim_{x \rightarrow 0} \frac{2^x + Ax - 1}{x^2}$$

exists and is finite. For this value of A , what is the limit equal to?

(c) Find a condition on the constant B that guarantees that the limit

$$\lim_{x \rightarrow 0} \frac{x^2 + x^B}{\ln(1+x) - x}$$

exists and is finite. For such a value of B , what is the limit equal to?

Problem 2

Keywords: Mistakes in L'Hôpital's rule

Your friend incorrectly computed two limits using L'Hôpital's rule. Their computations are included below. Find where they made a mistake, and provide the correct solution.

(a)

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{x^3}{e^x - x - 1} &\stackrel{(L'H)}{=} \lim_{x \rightarrow 0} \frac{3x^2}{e^x - 1} \\ &\stackrel{(L'H)}{=} \lim_{x \rightarrow 0} \frac{6x}{e^x} \\ &\stackrel{(L'H)}{=} \lim_{x \rightarrow 0} \frac{6}{e^x} \\ &= 6. \end{aligned}$$

(b)

$$\begin{aligned} \lim_{x \rightarrow \infty} \frac{x^2 + \cos(x)}{x^2} &\stackrel{(L'H)}{=} \lim_{x \rightarrow \infty} \frac{2x - \sin(x)}{2x} \\ &\stackrel{(L'H)}{=} \lim_{x \rightarrow \infty} \frac{2 - \cos(x)}{2} \end{aligned}$$

and since $\lim_{x \rightarrow \infty} \frac{2 - \cos(x)}{2}$ does not exist, neither does $\lim_{x \rightarrow \infty} \frac{x^2 + \cos(x)}{x^2}.$

Problem 3

Keywords: Extreme Values, Optimization

A powerhouse is on one edge of a straight river and a factory is on the other edge, 40 meters downstream. The river is 20 meters wide. We want to run an electric cable to connect the powerhouse to the factory. It costs $\$5\sqrt{2}$ per meter to run electrical cable across the river and \$5 per meter on land. We want to find how to install the cable to minimize the cost.

- (a) Draw a picture and introduce variable(s) to describe the situation.
- (b) Express the objective function and find the interval of interest.
- (c) How should we install the cable to minimize the cost?

Optional Problem *(just for fun, no need to submit)*

Let f be a function such that f , f' , f'' , and f''' are defined for all real numbers. Assume that

$$\lim_{x \rightarrow 0} f(x) = 0, \quad \lim_{x \rightarrow 0} f'(x) = 0, \quad \lim_{x \rightarrow 0} f''(x) = 0, \quad \text{and} \quad \lim_{x \rightarrow 0} f'''(x) = 4.$$

Find the following limits, if possible. If a limit does not exist or there is not enough information to determine the limit, explain why this is the case. If you use L'Hôpital's rule, make sure to indicate where and say which indeterminate form allows you to use it.

- (a) $\lim_{x \rightarrow 0} \frac{f(x)}{\sin x}$
- (b) $\lim_{x \rightarrow 0} \frac{f(x)}{3x - \sin(3x)}$
- (c) $\lim_{x \rightarrow 0^+} (1 + 5f''(x))^{1/x}$